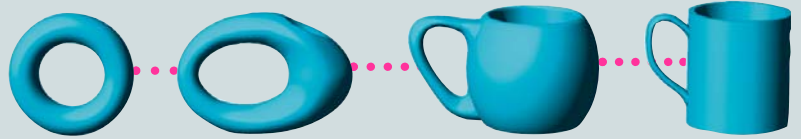


SHAPES that STRETCH

3D shapes can be topologically equivalent, too. A coin and a marble are topologically equivalent to each other, for instance, but neither is equivalent to a doughnut, which has a hole in the middle. However, a cup *is* equivalent to a doughnut, because both have a single hole going all the way through.



THE MÖBIUS STRIP

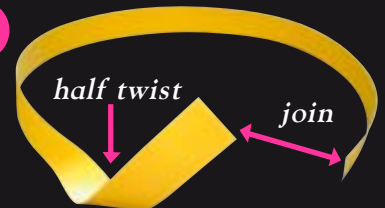
How many sides does a piece of paper have? Two, obviously. Is it possible for a piece of paper to have only one side? Yes, but it's a very strange piece of paper, invented by a mathematician called Möbius. This is how to make a Möbius strip.

1

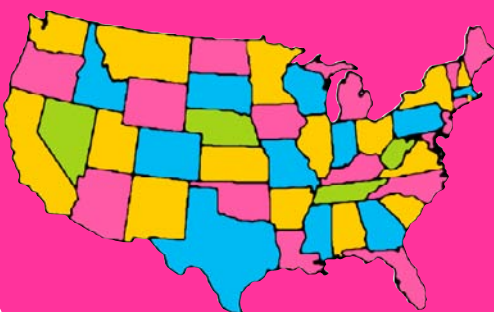


Neatly cut a long strip of paper. It should be at least 20 cm (8 in) long and at least 2.5 cm (1 in) wide.

2



Make a half twist and tape the ends together. The paper now has only one side and one edge! Run a finger around it to check.



The four-colour puzzle

What's the minimum number of colours you need to colour in a map so that neighbouring areas never have the same colour? This puzzle, posed by Möbius in 1840, baffled everyone until it was solved in 1976 by a computer, which took 1200 hours. Yet the puzzle seems so simple. You can try the puzzle for yourself. Draw a continent and divide it into countries. Make the sea around it one colour. Then see if you can colour in all the countries around the coast in only three colours.